

# Tangential Eigenvalue Universality on Constant-Curvature Surfaces

Sandbox for Paper I, Item #2

April 15, 2026

## Main result

**Theorem 1** (Riemannian Havelock universality). *On the unit 2-sphere, the tangential eigenvalue of the constrained Hessian  $\nabla^2(H_{\text{sph}} - \Omega J)$  at mode  $m \in \{1, \dots, N-1\}$  is*

$$\hat{e}_t^T \nabla^2(H_{\text{sph}} - \Omega J) \hat{e}_t = \frac{m(N-m)}{2\xi},$$

*identical to the flat-plane result at ring radius  $\sqrt{\xi}$ . The same holds on the hyperbolic plane with  $(1+\xi)$  replaced by  $(1-\xi)$  throughout.*

*Proof.* Decompose  $H_{\text{sph}} = H_{\text{flat}} + W$  where  $H_{\text{flat}} = -\sum_{j < k} \ln |z_j - z_k|$  and  $W = \frac{N-1}{2} \sum_k \ln(1+|z_k|^2)$ . The Lagrangian is  $\mathcal{L} = H_{\text{flat}} + W - \Omega J$  with  $J = \sum_k (1-|z_k|^2)/(1+|z_k|^2)$ . The tangential eigenvalue decomposes as

$$\hat{e}_t^T \nabla^2 \mathcal{L} \hat{e}_t = \underbrace{\hat{e}_t^T \nabla^2 H_{\text{flat}} \hat{e}_t}_{\text{flat Havelock}} + \underbrace{\hat{e}_t^T \nabla^2 (W - \Omega J) \hat{e}_t}_{\text{curvature + constraint}}.$$

We show the second term equals  $(N-1)/(2\xi)$ , i.e. the flat-plane Lagrange shift  $-2\mu_L = (N-1)/(2\xi)$  at ring radius  $\sqrt{\xi}$ . Combined with the flat-plane unshifted tangential eigenvalue  $(m(N-m) - (N-1))/(2\xi)$  (derived in the constrained-minimum proof, §6.2), this gives  $m(N-m)/(2\xi)$ .

**Step 1: Tangential projection of  $\nabla^2 W$ .** At vertex  $k$  on the ring ( $|z_k|^2 = \xi$ , angle  $\theta_k$ ), the Hessian of  $W_k = \frac{N-1}{2} \ln(1+|z_k|^2)$  has entries

$$\begin{aligned} \frac{\partial^2 W_k}{\partial x_k^2} &= \frac{N-1}{1+\xi} - \frac{2(N-1)\xi \cos^2 \theta_k}{(1+\xi)^2}, & \frac{\partial^2 W_k}{\partial y_k^2} &= \frac{N-1}{1+\xi} - \frac{2(N-1)\xi \sin^2 \theta_k}{(1+\xi)^2}, \\ \frac{\partial^2 W_k}{\partial x_k \partial y_k} &= -\frac{2(N-1)\xi \sin \theta_k \cos \theta_k}{(1+\xi)^2}. \end{aligned}$$

The mode- $m$  tangential unit vector at vertex  $k$  is  $\hat{e}_{t,k} \propto \alpha_k (-\sin \theta_k, \cos \theta_k)$  with  $\alpha_k = \cos(2\pi mk/N)$ . The tangential projection at vertex  $k$  is

$$\begin{aligned} \sin^2 \theta_k \frac{\partial^2 W_k}{\partial x_k^2} - 2 \sin \theta_k \cos \theta_k \frac{\partial^2 W_k}{\partial x_k \partial y_k} + \cos^2 \theta_k \frac{\partial^2 W_k}{\partial y_k^2} \\ = \frac{N-1}{1+\xi} - \frac{2(N-1)\xi}{(1+\xi)^2} [\sin^2 \theta_k \cos^2 \theta_k - 2 \sin^2 \theta_k \cos^2 \theta_k + \cos^2 \theta_k \sin^2 \theta_k] \\ = \frac{N-1}{1+\xi}. \end{aligned}$$

The bracketed expression vanishes identically ( $1 - 2 + 1 = 0$ ), so the projection is  $\theta_k$ -independent and hence *mode-independent*. Summing over vertices with  $\sum_k \alpha_k^2 = N/2 = \|\hat{e}_t\|^2$ :

$$\hat{e}_t^T \nabla^2 W \hat{e}_t = \frac{N-1}{1+\xi}. \quad (1)$$

**Step 2: Tangential projection of  $\nabla^2 J$ .** With  $J_k = (1 - |z_k|^2)/(1 + |z_k|^2)$ :

$$\begin{aligned} \frac{\partial^2 J_k}{\partial x_k^2} &= -\frac{4}{(1+\xi)^2} + \frac{16\xi \cos^2 \theta_k}{(1+\xi)^3}, & \frac{\partial^2 J_k}{\partial y_k^2} &= -\frac{4}{(1+\xi)^2} + \frac{16\xi \sin^2 \theta_k}{(1+\xi)^3}, \\ \frac{\partial^2 J_k}{\partial x_k \partial y_k} &= \frac{16\xi \sin \theta_k \cos \theta_k}{(1+\xi)^3}. \end{aligned}$$

The same tangential projection yields

$$-\frac{4}{(1+\xi)^2} + \frac{16\xi}{(1+\xi)^3} [\sin^2 \theta \cos^2 \theta - 2 \sin^2 \theta \cos^2 \theta + \cos^2 \theta \sin^2 \theta] = -\frac{4}{(1+\xi)^2},$$

again  $\theta$ -independent. Therefore

$$\hat{e}_t^T \nabla^2 J \hat{e}_t = -\frac{4}{(1+\xi)^2}. \quad (2)$$

**Step 3: Combining  $W - \Omega J$ .** With  $\Omega = (N-1)(1-\xi^2)/(8\xi) = (N-1)(1-\xi)(1+\xi)/(8\xi)$ :

$$\begin{aligned} \hat{e}_t^T \nabla^2 (W - \Omega J) \hat{e}_t &= \frac{N-1}{1+\xi} + \frac{(N-1)(1-\xi)(1+\xi)}{8\xi} \cdot \frac{4}{(1+\xi)^2} \\ &= \frac{N-1}{1+\xi} + \frac{(N-1)(1-\xi)}{2\xi(1+\xi)} \\ &= \frac{N-1}{1+\xi} \left[ 1 + \frac{1-\xi}{2\xi} \right] = \frac{N-1}{1+\xi} \cdot \frac{1+\xi}{2\xi} = \frac{N-1}{2\xi}. \end{aligned}$$

This equals exactly  $-2\mu_L = (N-1)/(2\xi)$ , the flat-plane Lagrange shift at ring radius  $\sqrt{\xi}$ .

**Step 4: Assembly.** The flat-plane unshifted tangential eigenvalue at radius  $\sqrt{\xi}$  is  $(m(N-m) - (N-1))/(2\xi)$  (from the constrained-minimum derivation). Adding the curvature-plus-constraint contribution  $(N-1)/(2\xi)$ :

$$\hat{e}_t^T \nabla^2 \mathcal{L} \hat{e}_t = \frac{m(N-m) - (N-1)}{2\xi} + \frac{N-1}{2\xi} = \frac{m(N-m)}{2\xi}.$$

**Hyperbolic case.** On  $\mathbf{H}^2$  with  $W = -(N-1)/2 \sum_k \ln(1 - |z_k|^2)$  and  $J = \sum_k (1 + |z_k|^2)/(1 - |z_k|^2)$  (the Poincaré-disk formulation), the same tangential-projection argument gives  $\hat{e}_t^T \nabla^2 W \hat{e}_t = (N-1)/(1-\xi)$  and  $\hat{e}_t^T \nabla^2 J \hat{e}_t = -4/(1-\xi)^2$ , and with  $\Omega_{\mathbf{H}^2} = -(N-1)(1+\xi)/(8\xi)(1-\xi)$  the combination again equals  $(N-1)/(2\xi)$ .  $\square$

## 1 Numerical verification

All intermediate identities are verified in `numerical_check.py` (this directory):

- $\hat{e}_t^T \nabla^2 W \hat{e}_t = (N-1)/(1+\xi)$ : confirmed numerically for  $N \in \{5, 6, 7\}$ ,  $\xi \in \{0.2, 0.5\}$ , all modes  $m \in \{2, \dots, N-2\}$  (mode-independent to  $10^{-2}$ ).
- $\hat{e}_t^T \nabla^2 J \hat{e}_t = -4/(1+\xi)^2$ : same parameters, mode-independent to  $10^{-2}$ .

- $\hat{e}_t^T \nabla^2(W - \Omega J) \hat{e}_t = (N - 1)/(2\xi)$ : confirmed for  $N \in \{5, 6, 7\}$ ,  $\xi \in \{0.15, 0.3, 0.5, 0.7\}$ .
- Full tangential eigenvalue  $= m(N - m)/(2\xi)$ : confirmed for  $N \in \{4, 5, 6, 7\}$ ,  $\xi \in \{0.15, 0.3, 0.5, 0.7\}$ , all modes.
- Symbolic: the cross-term cancellation  $\sin^2\theta \cos^2\theta - 2 \sin^2\theta \cos^2\theta + \cos^2\theta \sin^2\theta = 0$  is verified by `sympy` in the explicit tangential projection of  $\nabla^2 W$ .